Trigonometric Systems, Multiple-Angle Formulas, and a Determinant Recurrence

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§1.1 Correcting a trigonometric mistake

The first handwritten line says that an earlier calculation from February 20 contained a mistake. The attempted determinant manipulation involved vectors built from angles α and β, and a factor such as $\tan(\alpha - \beta)$. The note records that the determinant-style computation did not work and that the reason for the error was not immediately clear.

This is worth preserving. A failed determinant manipulation is not useless; it tells us that the problem probably has hidden dependencies among the trigonometric variables. Before choosing a linear algebra method, one must check whether the equations are genuinely linear in the chosen unknowns.

§1.2 A trigonometric system

The main problem asks for $\cos\alpha\cos\beta$ under conditions of the form

$$\sin\alpha + \sin\beta = \frac{1}{2}, \qquad \cos\alpha - \cos\beta = \frac{1}{3}.$$

The note introduces

$$a = \sin\alpha, \qquad b = \sin\beta, \qquad c = \cos\alpha, \qquad d = \cos\beta.$$

Then the equations become

$$a + b = \frac{1}{2}, \qquad c - d = \frac{1}{3}, \qquad a^2 + c^2 = 1, \qquad b^2 + d^2 = 1.$$

This transforms a trigonometric problem into an algebraic system.

§1.3 A rational-parametrization idea

The handwritten solution tries to express the four variables as sums of rational parts and radicals:

$$a = q_1 + r_1, \qquad b = q_2 - r_1, \qquad c = q_3 + r_2, \qquad d = q_4 + r_2,$$

with

$$q_1 + q_2 = \frac{1}{2}, \qquad q_3 - q_4 = \frac{1}{3}.$$

The two circle equations become quadratic equations in the radical parts. Comparing rational and irrational parts can force relations among the q_i and r_i.

The note finds relations such as

$$q_1^2 + q_3^2 = q_2^2 + q_4^2,$$

and a linear relation between r_1 and r_2. One conclusion written on the page is that the signs may be controlled by introducing one radical and then solving back for c, d. The method is experimental, and the note explicitly says that another method was not successful.

§1.4 A direct algebraic route

A cleaner derivation is to use sum-to-product formulas. We have

$$\sin\alpha + \sin\beta = 2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2} = \frac{1}{2},$$

and

$$\cos\alpha - \cos\beta = -2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2} = \frac{1}{3}.$$

Dividing gives

$$-\tan\frac{\alpha-\beta}{2} = \frac{2}{3},$$

provided the common factor is nonzero. This determines the half-difference angle. Then one can recover products such as

$$\cos\alpha\cos\beta = \frac{\cos(\alpha+\beta)+\cos(\alpha-\beta)}{2}.$$

The direct method shows why the algebraic system has hidden trigonometric structure.

§1.5 Multiple-angle formulas

The second page records standard multiple-angle formulas. One form is

$$\sin(nx) = \sum_{k=0}^{n}\binom{n}{k}\cos^k x\,\sin^{n-k} x\,\sin\left(\frac{(n-k)\pi}{2}\right),$$

and

$$\cos(nx) = \sum_{k=0}^{n}\binom{n}{k}\cos^k x\,\sin^{n-k} x\,\cos\left(\frac{(n-k)\pi}{2}\right).$$

These come from expanding

$$(\cos x + i\sin x)^n = \cos(nx) + i\sin(nx).$$

Thus De Moivre's formula is the conceptual source of the binomial-looking expressions.

The recurrence formulas are

$$\sin(nx) = 2\sin((n-1)x)\cos x - \sin((n-2)x),$$

$$\cos(nx) = 2\cos((n-1)x)\cos x - \cos((n-2)x),$$

and

$$\tan(nx) = \frac{\tan((n-1)x)+\tan x}{1-\tan((n-1)x)\tan x}.$$

The sine and cosine recurrences are Chebyshev-type recurrences.

§1.6 The determinant recurrence

The final page asks one to prove a determinant formula. The displayed matrix is tridiagonal, with diagonal entries involving $2\cos\theta$ and off-diagonal entries equal to 1, with a modified first entry involving $\cos\theta$. The intended proof is expansion along the last row or last column.

If D_n denotes the determinant of the $n \times n$ tridiagonal matrix with diagonal $2\cos\theta$ and off-diagonal 1, then it satisfies

$$D_n = 2\cos\theta\, D_{n-1} - D_{n-2}.$$

This is the same recurrence as the multiple-angle formulas. Therefore D_n can be expressed in terms of $\sin((n+1)\theta)/\sin\theta$ or a related Chebyshev polynomial, depending on the exact boundary convention.

The note's blue comment says that proving the determinant is a cosine expression can be done by the recurrence. This is exactly the right idea: the matrix determinant and the trigonometric formula are the same recurrence with the same initial values.

§1.7 From technique to structure

This note is about recognizing when a problem should be treated as trigonometry, algebra, or linear algebra. A failed determinant trick reveals hidden constraints. Sum-to-product formulas solve the angle system more naturally. Multiple-angle identities come from De Moivre's formula. Tridiagonal determinants satisfy the same recurrence as Chebyshev polynomials. The common theme is that recurrence relations are a bridge between trigonometry and matrices.

§1.8 Trigonometric systems as algebraic systems

A trigonometric system can often be converted into algebra by setting

$$u = \cos x, \qquad v = \sin x, \qquad u^2 + v^2 = 1.$$

This turns identities into polynomial relations on the unit circle. The constraint must always be kept; otherwise extraneous algebraic solutions appear.

§1.9 Rational parametrization of the circle

The substitution

$$t = \tan\frac{x}{2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad \sin x = \frac{2t}{1+t^2}$$

rationally parametrizes the unit circle except one point. It turns many trigonometric equations into rational equations.

§1.10 Multiple-angle formulas and Chebyshev polynomials

The identity

$$\cos(nx) = T_n(\cos x)$$

defines the Chebyshev polynomial T_n. Thus multiple-angle formulas are polynomial dynamics on the interval $[-1, 1]$.

§1.11 Determinant recurrences

A determinant recurrence often comes from expanding along a row or column. The resulting sequence is commonly governed by a second-order linear recurrence, whose characteristic roots explain closed forms.

§1.12 Trigonometry through algebraic parametrization

The note joins trigonometry and algebra: systems become polynomial constraints, half-angle substitution gives rational parametrization, multiple angles give Chebyshev polynomials, and determinant recurrences reveal linear recurrence structure.